

Control input

Passive components $\rightarrow$ Control input $\rightarrow$ Passive elements have a fixed output characteristics for a given state of

Active " $\rightarrow$ No " " " " Their char. cannot be controlled.

Diode $\rightarrow$ Current Fixed for certain voltage

Transistor $\rightarrow$ V can be controlled using gate/bases.

Active element $\rightarrow$ Can supply energy, amplify signals
control current flow in a ckt.

Point contact transistor $\rightarrow$ 1947

$\uparrow$ (BJT)
Bell Labs

MOS Trans $\rightarrow$ 1960

$\rightarrow$ Transistor as current source $\rightarrow$ Transistor as switch
Difference between analog and digital
 $\downarrow$ analog signals $\downarrow$ digital signals
 $\downarrow$ It is summation of sine waves $\downarrow$ 1/0

12 metal layers 2000 process steps

Water to circuit

How many process steps? $\rightarrow$ hardly 10 process steps.

Clean Rooms $\rightarrow$ 200-300 times filtering done in clean rooms.

PCB $\rightarrow$ made using fiberglass (insulator) } Definition:
IC

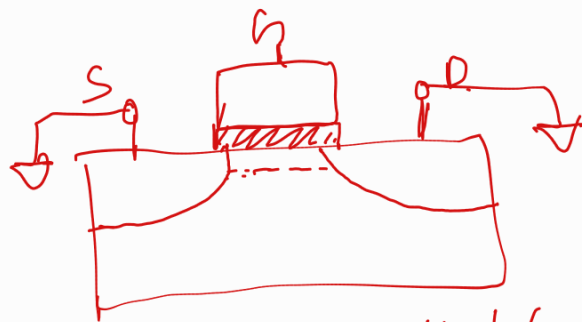
PCB $\rightarrow$ Printed circuit board is a flat board made using insulating material (fiberglass) It has etched/printed conductive paths which connect and mechanically support various comp.

IC $\rightarrow$ Integrated Circuits are miniaturized ckt in which active/passive elements are fabricated and interconnected in a single semiconductor chip. (usually Si) to perform, processing, memory, amp, etc.

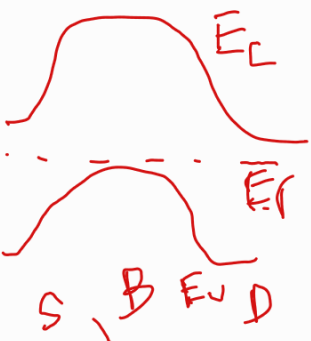
MLC/AI > Smartphone > Automotive > IoT

100^{years} FET

MOSFET



V_{gs} +ve. holes are repelled by $\vec{E}$ creating a region of -ve charges near SiO_2 and Si



As the $V_{int} > V_{th}$ the EB leads the $\vec{E}$ from S and D rush in to interface of Si/SiO_2

Inversion channel is formed

e^- conc & V_{ds} linear.

Steps of Fab

- (1) - Optical lithography
- (2) Thermal Oxidatⁿ
- (3) Deposition.
- (4) Etch
- (5) Ion implantation
- (6) Anneal
- (7) Silicidation
- (8) Copper plating

OTD EIA SC

Optical lithography
Thermal Oxidation.
Deposition.

Etch

Ion

Annealing

Silicidation

Copper pl^o

CMOS $\rightarrow$ 2000 steps 12 metal layers

Steps of IGP

- (1) Idea
- (2) Design
- (3) Wafer Fabricatⁿ
- (4) Packagⁱ
- (5) Sales.

IDM Integrated Device Manufacturer

Foundries

Fabless

IDM IC Design Tech Dev Fabric. Sales

Foundries

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|,

Fabless

IC

Sales

More and more Fabless

Semiconductor ecosystem is an umbrella of
many companies who work together
to produce an IC.

OTD

EIA

SC

Steps for IC

IC des

Fab

Pack

Sales/D.

Semiconductor

IP →

EDA

Chem/Wafer.

Wafer Fab equipment

Fabless

IDM.

Found

IP $\rightarrow$ They sell their chip design or software building (IP core)
arm, syn, cadence.

EDA $\rightarrow$ Electronic Design Autom - Cadence, Siemens, Syn
Mentor

Chem/Water $\rightarrow$ si, he, SOI, GaN, GaAs
Linde, Global Waters,

Water Fabry $\rightarrow$ EUV tool, Fab tools
Appl, Lam, ASML

Fabless

IDM $\rightarrow$ Samsung Intel Texas
Intr, Samsung, yf

Outsourced Semiconductor Assembly and Test (OSAT)
Chip MOS / Power tech